1 **(a)** $12x^{-3}$ or $\dfrac{12}{x^3}$ **[B1]**

(b) $\dfrac{2x^3}{3}$ or $\dfrac{2}{3}x^3$ **[B2]**

Correct cube root of coefficient $\dfrac{2}{3}$ OR correct index x^3 seen **[M1]**

[Total Marks: 3]

2 **(a)** $20x$ **[B1]**

(b) Numerator $4x^8$ or denominator y^6 seen **[M1]**

$\dfrac{4x^8}{y^6}$ **[A1]**

(c) -3 **[B1]**

[Total Marks: 4]

3 **(a)** w^{11} **[B1]**

(b) 3^2 or 9 seen, or x^6 seen **[M1]**

$9x^6$ **[A1]**

(c) $\dfrac{15}{5} = 3$ or $y^{8-(-2)}$ **[M1]**

$3y^{10}$ **[A1]**

[Total Marks: 5]

4 **(a)** p^{11} **[B1]**

(b) $9q^6$ **[B2]**

Partial correct answer $9q^k$ or cq^6 **[B1]**

(c) $(\sqrt[3]{27})^2$ or 3^2 or $\sqrt[3]{729}$ **[M1]**

9 **[A1]**

[Total Marks: 5]

5 **(a)** $\dfrac{1}{4}$ or 0.25 **[B2]**

Correct intermediate step e.g., 2^{-2} or $\left(\dfrac{1}{32}\right)^{\frac{2}{5}}$ or $32^{\frac{1}{5}} = 2$ **[M1]**

(b) Writes 125 as 5^3 **[M1]**

Equates indices: $x + 2 = 3x$ **[M1]**

1 **[A1]**

[Total Marks: 5]

6 **(a)** $\frac{1}{4}$ or 0.25 **[B1]**

(b) 25 or $\frac{1}{25}$ seen **[M1]**

1 **[A1]**

(c) $125 = 5^3$ seen or implied **[B1]**

$2x - 1 = 3x$ **[M1]**

$x = -1$ **[A1]**

[Total Marks: 6]

7 **(a)** $4 \times \frac{3}{2} = 6$ or $x^{-3+\frac{1}{2}}$ **[M1]**

$6x^{-\frac{5}{2}}$ or $\frac{6}{x^{\frac{5}{2}}}$ **[A1]**

(b) $3^{2x+1} = 3^3$ or $2x + 1 = 3$ **[M1]**

1 **[A1]**

(c) $8^{\frac{1}{3}} = 2$ or $(a^6)^{\frac{1}{3}} = a^2$ or $(b^3)^{\frac{1}{3}} = b$ **[M1]**

$\frac{2a^2}{b}$ **[A1]**

[Total Marks: 6]

8 **(a)** 8 **[B1]**

x^9 **[B1]**

y^6 as denominator or y^{-6} **[B1]**

(b) $8 = 2^3$ seen or implied by $8^x = 2^{3x}$ **[M1]**

$x + 3 = 3x$ **[M1]**

1.5 or $\frac{3}{2}$ **[A1]**

[Total Marks: 6]

9 **(a)** Expressing both sides with base 2: $(2^4)^{2x-1} = (2^3)^{x+2}$ **[M1]**

Equating powers: $4(2x - 1) = 3(x + 2)$ or $8x - 4 = 3x + 6$ **[M1]**

2 **[A1]**

(b) Taking the reciprocal to remove negative index: $\left(\frac{16y^8}{81x^{12}}\right)^{\frac{3}{4}}$ **[M1]**

Evaluating the fractional power of the numbers: $16^{\frac{3}{4}} = 8$ and $81^{\frac{3}{4}} = 27$ **[M1]**

$\frac{8y^6}{27x^9}$ **[A1]**

[Total Marks: 6]

10 **(a)** $12x^{\frac{3}{2}}$ seen as numerator **[M1]**

$2x^k$ for any k or $cx^{\frac{7}{6}}$ for any c **[M1]**

$2x^{\frac{7}{6}}$ **[A1]**

(b) Converting all terms to base 3: $3^{2(2x-1)} \times 3^{x+1} = 3^{3x}$ **[M1]**

Applying exponent multiplication law: $3^{4x-2+x+1}$ or 3^{5x-1} **[M1]**

Equating indices: $5x - 1 = 3x$ **[M1]**

0.5 or $\frac{1}{2}$ **[A1]**

[Total Marks: 7]

11 **(a)** $4x^2$ or y^{-3} or $\frac{1}{y^3}$ seen **[M1]**

$\frac{4x^2}{y^3}$ or $4x^2y^{-3}$ **[A1]**

(b) 3^{2x} or $3^{3(x-1)}$ seen **[B1]**

$3\sqrt{3} = 3^{1.5}$ or $3^{\frac{3}{2}}$ seen **[B1]**

$2x + 3x - 3 = 1.5$ or equivalent equation in x **[M1]**

$x = 0.9$ or $\frac{9}{10}$ **[A1]**

[Total Marks: 6]

12 **(a)** $3x^2y^{-6}$ or $\frac{3x^2}{y^6}$ **[B3]**

Coefficient 3 seen **[M1]**

Correct subtraction of indices $x^{\frac{3}{2}-\left(-\frac{1}{2}\right)}$ leading to x^2 OR y^{-2-4} leading to y^{-6} **[M1]**

(b) Expresses all terms as powers of 2: $(2^2)^{2x-1} \times (2^3)^{x+1} = (2^5)^x$ **[M1]**

Correctly adds powers on the LHS: $2(2x-1) + 3(x+1)$ or $4x - 2 + 3x + 3$ **[M1]**

Forms linear equation equating LHS and RHS powers: $7x + 1 = 5x$ **[M1]**

$-\frac{1}{2}$ or -0.5 **[A1]**

[Total Marks: 7]

13 **(a)** -3 **[B1]**

(b) $2y^3$ **[B2]**

If not B2, then B1 for 2 or y^3 seen **[B1]**

(c) $\frac{1}{8}$ **[B1]**

[Total Marks: 4]

14 **(a)** $4x^3y^4$ **[B2]**

If not B2, then B1 for two of 4, x^3, y^4 correct **[B1]**

(b) $(3^3)^{x-1} = (3^2)^{2x+1}$ or $3^{3(x-1)} = 3^{2(2x+1)}$ **[M1]**

$3x - 3 = 4x + 2$ **[M1]**

-5 **[A1]**

[Total Marks: 5]

15 **(a)** $\left(\frac{27y^6}{8x^9}\right)^{\frac{2}{3}}$ or $\frac{4x^6}{9y^4}$ (inverted or squared/cubed root correctly applied to some terms) **[M1]**

$\left(\frac{3y^2}{2x^3}\right)^2$ **[M1]**

$\frac{9y^4}{4x^6}$ or $2.25y^4x^{-6}$ **[A1]**

(b) $\frac{5^{a-2} \times 5^{2b}}{5^{3(a+1)}} = 5^4$ (expressing all terms as powers of 5) **[M1]**

$a - 2 + 2b - 3(a + 1) = 4$ **[M1]**

$-2a + 2b - 5 = 4$ or $-2a + 2b = 9$ **[A1]**

Accept any pair of values satisfying the equation, e.g. $a = 0, b = 4.5$ or $a = -4.5, b = 0$ or $b = a + 4.5$ **[A1]**

[Total Marks: 7]

16 **(a)** 15 or x^2 in a product **[M1]**

$15x^2$ **[A1]**

(b) $\sqrt[4]{16}$ or 2^3 or $4096^{\frac{1}{4}}$ **[M1]**

8 **[A1]**

[Total Marks: 4]

17 **(a)** Numerator $8x^{15}$ **[B1]**

Denominator $27y^6$ **[B1]**

$\frac{8x^{15}}{27y^6}$ **[A1]**

(b) Writing both sides with a common base of 3, e.g. $3^{2x} = 3^{3(x+1)}$ **[M1]**

$2x = 3x + 3$ **[M1]**

-3 **[A1]**

[Total Marks: 6]

18 (a) Expressing all terms with a base of 2, e.g. $2^{2x} \times 2^{3(2x-1)} = 2^4$ **[M1]**

Correct linear equation for the exponents: $2x + 6x - 3 = 4$ or $8x = 7$ **[M1]**

$\frac{7}{8}$ or 0.875 **[A1]**

(b) Simplifying the numerator to $24x^1$ (or $24x$) **[M1]**

Correct manipulation of the coefficients $\frac{24}{8} = 3$ or indices $x^{1-(-3)}$ **[M1]**

$3x^4$ **[A1]**

[Total Marks: 6]

19 (a) y^3 **[B1]**

(b) $\frac{1}{25}$ or 0.04 **[B1]**

(c) $8x^{12}$ **[B2]**

If B0, allow [M1] for $8x^k$ or kx^{12} where $k \neq 0$

[Total Marks: 4]

20 (a) $(\sqrt[3]{125})^2$ or 5^2 or $125^2 = 15625$ and $\sqrt[3]{15625}$ **[M1]**

25 **[A1]**

(b) 3 **[B1]**

x^6 **[B1]**

y^{-8} or $\frac{1}{y^8}$ **[B1]**

Note: Award [B3] for a fully correct final answer of $\frac{3x^6}{y^8}$ or $3x^6y^{-8}$

[Total Marks: 5]

21 (a) $(3^2)^{2x} = (3^3)^{x+1}$ or $3^{4x} = 3^{3x+3}$ **[M1]**

$4x = 3x + 3$ **[M1]**

3 **[A1]**

(b) Inverts fraction to remove negative power: $\left(\frac{81b^{12}}{16a^8}\right)^{\frac{3}{4}}$ **[M1]**

Applies power $\frac{1}{4}$ (fourth root) to all terms: $\frac{3b^3}{2a^2}$ **[M1]**

Cubes the result to get final answer: $\frac{27b^9}{8a^6}$ or $3.375b^9a^{-6}$ **[A1]**

[Total Marks: 6]